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Block

3

QUANTITATIVE METHODS-I

Course Coordinator and Editor: **Prof. Narayan Prasad**

Block Preparation Team

Unit	Resource Person	IGNOU Faculty (Format, Language and Content Editing)
9	Dr. Anoop Chatterjee Associate Professor in Economics ARSD College (University of Delhi), Delhi	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi
10	Shri B.S. Bagla Associate Professor in Economics PGDAV College (University of Delhi), Delhi	Prof. Narayan Prasad Professor of Economics IGNOU, New Delhi
11	Prof P.K. Chaubey IIPA, New Delhi	Prof. Narayan Prasad Professor of Economics, IGNOU, New Delhi
12	Dr. Sunil K Mishra Fellow, Institute for Human Development New Delhi	Prof. Narayan Prasad Professor of Economics, IGNOU, New Delhi

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BLOCK 3 QUANTITATIVE METHODS-I

Economic theory is basically concerned about economic laws. These laws (or hypotheses) are qualitative statements on the economic behaviour of various agents both at the micro and macro levels. The validity of such qualitative statements, however, depends upon their rigorous empirical verification by means of the available data.

The implicit argument between the economic theory and its empirical verification is that if the stated laws holds good in the real world situation, it should be reflected in the displayed pattern of the relevant data. The identification and the examination of such patterns are conducted by employing various statistical techniques. The primary statistical tool employed for such a purpose is the *Regression Modeling*. Hence, a thorough knowledge of Regression Models is absolutely essential for conducting any serious research in Applied Economics.

An economic law or theory essentially postulates some relationship that exist among the economic variables. Suppose the relationship between two variables Q and P (i.e. Quantity and Price) is expressed in the form (say linear form) $Q = \alpha + \beta P$. Such an expression amount to an exact linear relationship which is hard to find in reality. Usually, various non-deterministic factors (called random factors) affect such a relationship. Therefore, in order to allow for the effect of such random factors, a random component is incorporated into the model i.e. $Q = \alpha + \beta P + U$, where U is a random variable. With this, the strict mathematical relationship expressed before becomes statistical in character. Regression technique can now be applied to estimate the two parameters α and β by using the actual data. This is the crux of a *Two Variable Regression Model* explained in **Unit 9**.

Unit 10 explains the multiple regression models by including more than two independent random variables. The same procedure as done in the case of two variable regression models is applied to examine whether the estimated parameters correspond to the postulated relationship. The treatment accorded to the Two Variable Regression Models and the Multiple Regression Models in this course is first limited to the ordinary least square (OLS) method subsequently, another method viz., the maximum likelihood method (MLM) is also explained. The two methods taken together equip a researcher with the basic tools for undertaking empirical research.

Improvement in well being of the poor has been one of the important goals of economic policy and to a significant extent it is determined by the growth and distribution of its income. Distribution patterns have an important bearing on the relationship between average income and poverty levels. Extreme inequalities are economically wasteful. Further, income inequalities also interact with other life-chance inequalities. Hence reducing inequalities has become priority of public policy. **Unit 11** deals with the conceptual and measurement aspects of income inequality.

The complex social and economic issues like child deprivation, food security, human development, human wellbeing etc. are difficult to measure in terms of single variable. They are expression of several indicators. Composite Index is an important statistical techniques to express the single value of several interdependent or independent variables. Hence, **Unit 12** throws light on various methods to construct Composite Index.

UNIT 9 TWO VARIABLE REGRESSION MODELS

Structure

9.0 OBJECTIVES

After reading this unit, you will be able to:

- know the issue of linearity in regression model;
- appreciate the probabilistic nature of the regression model;
- distinguish between the population regression model and the sample regression model;
- state the assumptions of the classical regression model;
- estimate the unknown population regression parameters with the help of the sample information;
- explain the concept of goodness of fit;
- use various functional forms in the estimation of the regression model;
- state the role of classical normal regression model; and
- conduct some tests of hypotheses regarding the unknown population regression parameters.

9.1 INTRODUCTION

The empirical research in economics is concerned with the statistical analysis of economic relations. Often these relations are expressed in the form of regression equations involving the dependent variable and the independent variables. The formulation of an economic relation in the form of a regression equation is called a regression model in econometrics. In such kind of a regression model, the variable under study is assumed to be a function of certain explanatory variables. The major purpose of a regression model is the estimation of its parameters. In addition, it is also useful for testing hypotheses and making certain forecasts. However, our concern will be mainly with the estimation of parameters and testing of some hypotheses. At this stage, we shall refrain from discussing the issue of forecasts from a regression model. A regression model that contains one explanatory variable is called a two variable regression model. In this unit, we shall focus on a two variable regression model. It may be mentioned here that we are essentially focusing on what is known as the two variable classical regression model.

9.2 THE ISSUE OF LINEARITY

When we talk about a regression model; at our level, what we have in our mind is a linear regression model. It is important to have a clear idea about the concept of linearity in the context of a regression model.

Suppose, we have a regression equation like,

$$Y = \alpha + \beta X$$

Generally, we call this a linear regression equation. By linearity we often mean a relationship, in which, the dependent variable is a linear function of the independent variable. In this case, the graphical representation of the regression equation is a straight line. However, there can be another interpretation of linearity. Consider the following regression equation,

$$Y = \alpha + \beta X + \gamma X^2$$

Now, this regression equation is linear in parameter, in the sense that the highest power of any parameter (constant) is one. But this equation is non-linear in variable because the highest power of the independent variable is two. In fact, conventionally speaking, it is a quadratic regression equation. In this way, we can have any number of polynomial regression equations with the highest power of the independent variable going up to any positive integer. And all these equations may be linear in parameter.

But consider this regression equation,

$$Y = \alpha + \beta^2 X$$

This regression equation is linear in variable, but non-linear in parameter because the power of β is two.

We should note now that from the point of view of regression analysis, we shall consider only those models which are linear in parameter, no matter whether they are linear in variable or not. After all, the main purpose of a regression model is the estimation of its parameters. These parameters have to be linear for the purpose of their straightforward estimation, say,

by least square method. For example, the parameters of the second regression equation presented above, can be easily estimated by a simple extension of the least square method that we have studied in Unit 8. It should be mentioned here that there are some regression equations that are non-linear in parameters. By applying suitable transformations they can be reduced to linear in parameter regression equations. We may consider the following regression equation

$$Y = \alpha X^{\beta}$$

This regression equation in its present form is non-linear in parameter. However, by applying log transformation, we obtain the following equation

$$\log Y = \log \alpha + \log \beta X$$

It can be clearly seen that the equation has now been transformed into one that is linear in the log of parameters. We can easily estimate log values of these parameters by using the usual least square method and then obtain the estimated values of the original parameters by applying the antilog procedure. However, some times, the regression equations can be of non-linear in parameter type that no transformation can render them to a linear in parameter form. Such equations should be the basis of non-linear or intrinsically non-regression models. There is no direct method available for the estimation of the parameters of this kind of regression models and they are generally estimated by following some standard iteration procedure. Any discussion on such iteration procedure, however, is beyond the scope of the present unit.

9.3 THE NON-DETERMINISTIC NATURE OF THE REGRESSION MODEL

We have already referred to the statistical nature of the regression equation in the last unit. By the very nature of social science, we cannot expect the relationships that may exist among different variables to be exact or deterministic. There is always some random element involved in them. As a result, for a particular value of the independent variable *X*, the value of the dependent variable *Y* cannot be exactly determined from such a relationship. Let us consider the example of a consumption function. Here, for a given level of income, when we try to identify the corresponding level of consumption, in all probability, we shall not be able to obtain a definite value of consumption; instead, a host of values will be available. And this will be the case with all possible levels of income. The diagram below shows this.

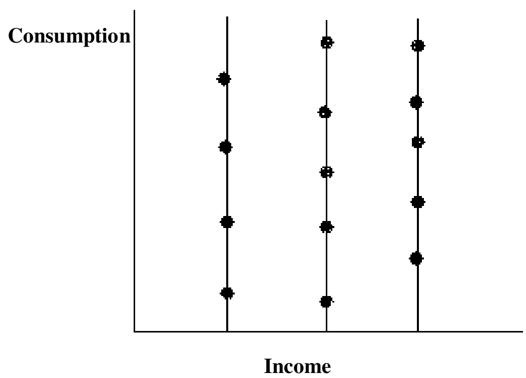

Fig. 9.1: Bi-Variate Population

Thus, the dependent variable Y tends to be probabilistic or stochastic in nature. In fact, for each value of the independent variable X, there will be a distribution of the values of the dependent variable Y. Accordingly; we specify the regression model by incorporating a random or stochastic variable. As mentioned in the previous unit, it should be clear that in our formulation the dependent variable is stochastic but the independent or explanatory variables are non-stochastic in nature. We should point out here that at an advanced treatment of the regression model, even the explanatory variables are considered to be stochastic in nature. It is the element of randomness either in the dependent variable alone or both in the dependent and independent variables, that make the regression model non-deterministic in nature.

9.4 POPULATION REGRESSION FUNCTION

The first step in the regression model is the conceptualization of a population regression function. Let us assume that there is a bi-variate population of (X, Y) of size N. Suppose, in this population, Y is the dependent variable and X is the independent variable. Let X and Y be related to each in a linear fashion in the following way:

$$Y = \alpha + \beta X + U$$

The above equation is called the population regression function. Here, α and β are the unknown parameters. It is clear from our discussion on linearity that this function is linear in variable as well as in parameter.

In this population regression function, the variable U deserves some attention. We have already mentioned that in social Science, there cannot be an exact relationship and it is at the most statistical in nature. In our formulation, in fact, Y is stochastic or random, but X is non-stochastic or deterministic in nature. Consequently, a random variable like U is introduced in the population regression function to incorporate this element of randomness of a statistical relationship. The random variable U is also called the disturbance term. It is a sort of catch – all variables that represent all kinds of indeterminacies of an inexact relationship. Thus it may represent, for example,

a) *Inherent randomness in human behaviour*: The unpredictability of human psyche is also reflected in the human behaviour. As a result, even after taking into account of all possible factors, a regression equation cannot fully explain the value of the dependent variable for the given values of all possible explanatory variables.

b) *Effect of omitted variables*: Sometimes for the sake of parsimony, all the explanatory variables are not included in a regression model. The disturbance term can be taken as a representative variable of all such omitted variables.

c) *Effect of measurement error*: Sometimes, it is not possible to measure the values of the dependent variable accurately. Consequently, the disturbance term is introduced in the regression model to represent the combined effect of all such possible sources of measurement error.

d) *Error in the formulation*: Often, the functional form of the regression model proves to be far from a correct depiction of the underlying relationship between the dependent variable and the independent variables.

1 We incorporate the disturbance term to correct the distortions that may arise from a wrong specification of the regression model.

We should note that apart from the above-mentioned factors, there might be other unidentifiable factors that might also be influencing the dependent variable in an unknown manner. We introduce the disturbance term for incorporating the effect of all such unknown random factors. It should be clear that for some of the reasons mentioned above the disturbance term may assume a positive value and for some other factor it may tend to be negative. Consequently, for all practical purposes, the net or the mean effect of the random disturbance can be taken to be zero.

Difference between the Disturbance Term and the Intercept

There is a difference between the interpretation of the disturbance term and that of the intercept term α in our regression equation. As we have noted, that the disturbance represents the random effects of the known and unknown variables that have not been included in the regression model. The intercept term, on the other hand, stands for all such variables, that are known to have some definite non-random effects on the dependent variable. For example, in the demand function, if we just formulate a two variable regression equation between the quantity demanded and the price instead of a multiple regression equation, then we are consciously not including variables like prices of other commodities and the income of the consumer. And all these variables affect the quantity demanded in a known fashion. So, the intercept term here represents the average effect of all such known factors that are not explicitly included in the model.

It should be clear now that it is the disturbance term that makes the relationship statistical in nature and renders the entire procedure so rich in content.

Population Regression Line

The main objective of the regression analysis is to obtain the value of the dependent variable for a given value of the independent variable. This can be attempted by running a regression on the population regression function $Y = \alpha + \beta X + U$, i.e., by fitting a regression line through the population cluster shown in Figure 9.1. Obviously, the value of Y that we shall obtain from such a population regression line will be an average value. In other words, we shall get the equation for the population regression line by applying the expectation to the population regression function, $Y = \alpha + \beta X + U$. Thus, the population regression equation will be given by

$$E(Y / X) = \alpha + \beta X$$

It may be mentioned here that while applying the expectation operator, E (U) can be taken as zero because, as we have discussed above, the mean effect of U tends to be zero.

Sometimes we call the above expression, the population regression line and loosely write it as

$$Y = \alpha + \beta X$$

In this expression, however, we should be clear that Y stands for the conditional mean of Y for a given X.

Assumptions of the Classical Regression Model

As we have mentioned earlier, the population regression model $Y = \alpha + \beta X + U$ is unknown in the sense that that the numerical values of its parameters are not known. As a result, we are not in a position to find the value taken by *Y* on an average for a given value of *X*. This means that we have to estimate these parameters from sample information. We already know that a very simple and popular method of estimation is the least square procedure. However, we should ensure that these least square estimates faithfully represent the unknown population parameters, otherwise, the whole purpose is defeated. This is possible only if the disturbance term *U* satisfies some assumptions. These assumptions along with two other that we have already mentioned about, are known as the assumptions of the classical regression model. The assumptions are:

1) The disturbance term *U* has a zero mean for all the values of *X*, i.e., $E(U) = 0$.
2) Variance of *U* is constant for all the values of X, i.e., $V(X) = \sigma^2$. It may be mentioned here that this assumption is known as the assumption of homoscedasticity in the econometric literature.
3) The disturbance terms for two different values of *X* are independent i.e., $\text{cov}(U_i, U_j) = 0$, for $i \neq j$.
4) *X* is non-stochastic.
5) The model is linear in parameter.

The assumption of non-stochasticity of *X* leads to a corollary. It is, *X* and *U* are independent i.e., $\text{cov}(X, U) = 0$.

9.5 SAMPLE REGRESSION FUNCTION

To put the whole discussion in the proper perspective; we have an unknown bi-variate population. The only information that we have is some randomly selected values of *Y* from this population that correspond to some fixed values of *X*. Suppose, in this way we have *n* pairs of values of *X* and *Y*. So, we can say that we have a random sample of size *n*. Our purpose is to estimate the parameters of this unknown population from our sample information so that we can have a reasonable estimate of an average value of *Y* for a given value of *X* that is valid for the entire population. In other words, our objective is to estimate the population regression function from the sample observations. We can proceed in that direction by discussing the concept of a sample regression function. The form of the sample regression function is quite similar to that of the population regression function. It can be presented as

$$\hat{Y} = \hat{\alpha} + \hat{\beta} X + \hat{U}$$

In the above expression, $\hat{Y}, \hat{\alpha}, \hat{\beta}$ and $\hat{U}$ should be interpreted as the sample estimators for their respective unknown population counterparts. Thus, we are hypothesizing that corresponding to the linear population regression function $Y = \alpha + \beta X + U$, there is a linear sample regression function $\hat{Y} = \hat{\alpha} + \hat{\beta} X + \hat{U}$. Our purpose is now to estimate the population regression function from the sample regression function. It means that we have to estimate $\hat{\alpha}$ and $\hat{\beta}$ from the given sample and take them as the estimates for the unknown population